

Skills Summary

Integrating Rational Functions with Partial Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Distinct Linear Factors

Setup. One constant per factor:

$$\frac{1}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}.$$

Clear denominators, then substitute $x = a$ and $x = b$ in turn — each choice kills one unknown. Each piece integrates to a logarithm of an absolute value.

Example: Evaluate $\int \frac{1}{(x-3)(x+1)} dx$.

Step 1. The denominator is already two distinct linear factors, so one constant over each is the whole decomposition.

Step 2. $1 = A(x+1) + B(x-3)$ gives $A = \frac{1}{4}$ at $x = 3$ and $B = -\frac{1}{4}$ at $x = -1$.

$$\Rightarrow \frac{\ln|x-3|}{4} - \frac{\ln|x+1|}{4} + C$$

Try it: Evaluate $\int \frac{1}{(x-5)(x+1)} dx$.

$$C + \frac{9}{|1+x|} - \frac{9}{|5-x|} \quad \text{check}$$

Skill 2 — Repeated and Quadratic Factors

Repeated factor: give it every power. $\frac{A}{x-a} + \frac{B}{(x-a)^2}$.

Irreducible quadratic: its numerator is *linear*. $\frac{Bx+C}{x^2+k}$, which integrates to a logarithm plus an arctangent. Counting unknowns is the check: they must equal the degree of the denominator.

Example: Evaluate $\int \frac{x+3}{(x+1)^2} dx$.

Step 1. The factor is repeated, so the decomposition needs both the first and the second power rather than a single term.

Step 2. $x+3 = A(x+1) + B$ gives $A = 1$ and $B = 2$, so the integral splits into $\int \frac{1}{x+1} dx + \int \frac{2}{(x+1)^2} dx$.

$$\Rightarrow \ln|x+1| - \frac{2}{x+1} + C$$

Try it: Evaluate $\int \frac{x+5}{(x+1)^2} dx$.

$$C + \frac{1+x}{4} - |1+x| \ln|x+1| \quad \text{check}$$

Skill 3 — Totals from a Rational Rate

Reading the application. If a rate is a rational function of time, the total accumulated is a definite integral — decompose, integrate to logarithms, then subtract at the limits.

Two logarithms combine: $\ln |f| - \ln |g| = \ln \left| \frac{f}{g} \right|$, which usually makes the arithmetic at the limits much shorter.

Example: Evaluate $\int_0^1 \frac{4}{(x+1)(x+3)} dx$.

Step 1. This is a total over an interval, so decompose first and only then substitute the limits into the combined logarithm.

Step 2. $\frac{2}{x+1} - \frac{2}{x+3}$ integrates to $2 \ln \frac{x+1}{x+3}$, and at the limits that is $2 \ln \frac{2}{4} - 2 \ln \frac{1}{3}$.
 $\Rightarrow 2 \ln \frac{3}{2}$

Try it: Evaluate $\int_0^3 \frac{4}{(x+1)(x+3)} dx$.

Check: $2 \ln 2$